

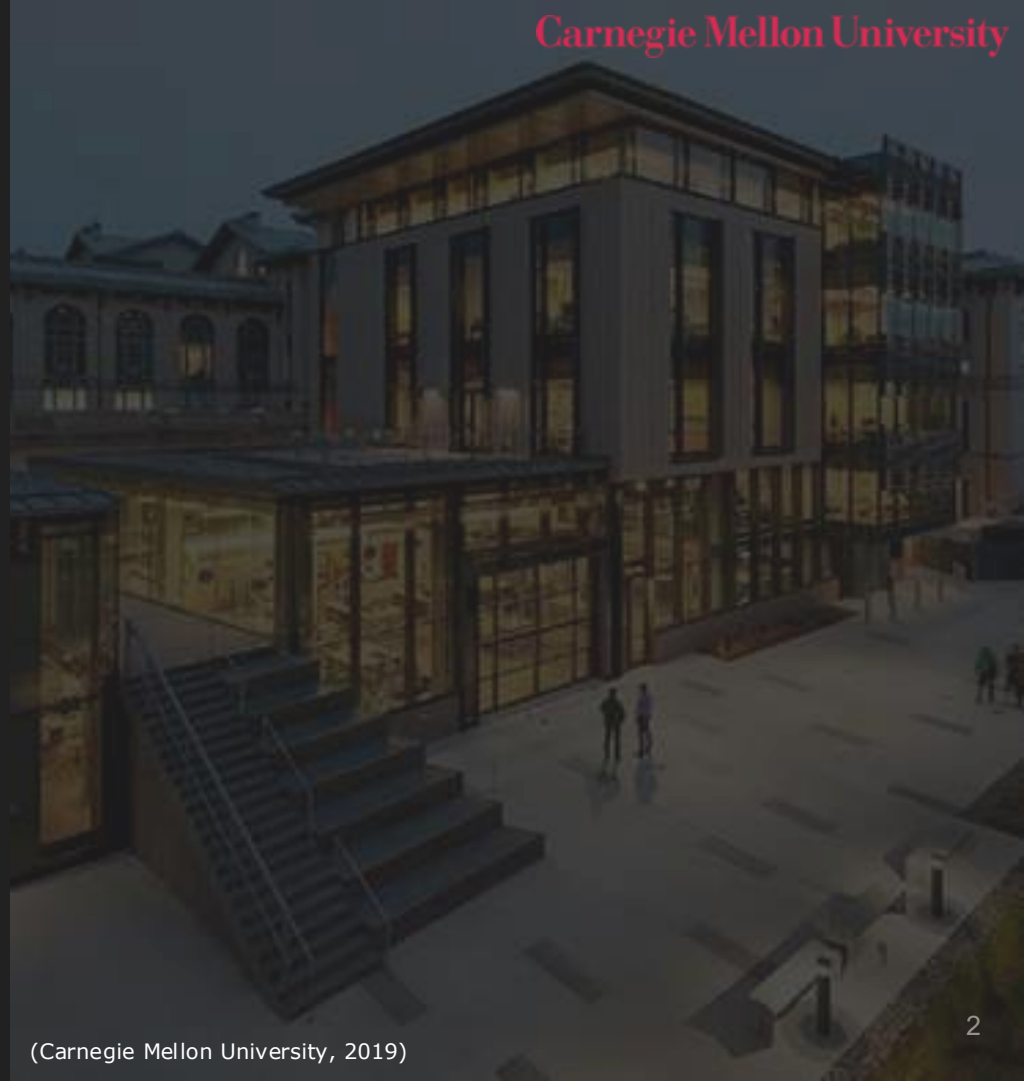
Re-Designing ANSYS Hall:

Manage Stormwater through Creating an Interactive Space

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Table of Contents

- 1. Problem Statement**
- 2. Objectives**
- 3. Methods of Calculations**
- 4. Five Individual Designs**
 - a. Overview**
 - b. Assessment**
- 5. Final Design**
 - a. Overview**
 - b. Assessment & Comparisons**
- 6. Conclusion**
- 7. Q&A**



The current **stormwater management system** of **ANSYS Hall** does not improve the rate or the volume of stormwater runoff compared to those of **pre-construction**.



***Following equations
were used to calculate
the rate and the volume
of stormwater runoff***

$$Q_y = C \times I \times A$$

Variables:

Q_y = peak runoff rate [cfs]

C = runoff coefficient of area [unitless]

I = average rainfall intensity [in/hr]

A = size of drainage area [acre]

Equations:

Given: $Q = \frac{(P - I_a)^2}{(P - I_a) + S}$

Empirical approximation: $I_a = 0.2S$

Further: $S = \frac{1000}{CN} - 10$

So: $Q = \frac{(P - 0.2S)^2}{(P + 0.8S)}$

To calculate a volume: $V = Q \times A$

Variables:

Q = runoff [inches]

P = rainfall [inches]

I_a = initial abstraction [inches]

The 'depth' of water which the site can manage before ponding/runoff occurs

S = potential maximum retention after runoff begins [inches]

Based on soil characteristics

CN = curve number [unitless]

A = site area [scale appropriately]

How Were the Uncertain Values Chosen?

Table 1 Runoff Coefficients for the Rational Method

	FLAT	ROLLING	HILLY
Pavement & Roofs	0.90	0.90	0.90
Earth Shoulders	0.50	0.50	0.50
Drives & Walks	0.75	0.80	0.85
Gravel Pavement	0.85	0.85	0.85
City Business Areas	0.80	0.85	0.85
Apartment Dwelling Areas	0.50	0.60	0.70
Light Residential: 1 to 3 units/acre	0.35	0.40	0.45
Normal Residential: 3 to 6 units/acre	0.50	0.55	0.60
Dense Residential: 6 to 15 units/acre	0.70	0.75	0.80
Lawns	0.17	0.22	0.35
Grass Shoulders	0.25	0.25	0.25
Side Slopes, Earth	0.60	0.60	0.60
Side Slopes, Turf	0.30	0.30	0.30
Median Areas, Turf	0.25	0.30	0.30
Cultivated Land, Clay & Loam	0.50	0.55	0.60
Cultivated Land, Sand & Gravel	0.25	0.30	0.35
Industrial Areas, Light	0.50	0.70	0.80
Industrial Areas, Heavy	0.60	0.80	0.90
Parks & Cemeteries	0.10	0.15	0.25
Playgrounds	0.20	0.25	0.30
Woodland & Forests	0.10	0.15	0.20
Meadows & Pasture Land	0.25	0.30	0.35
Unimproved Areas	0.10	0.20	0.30

Note:

- **Impervious surfaces in bold**
- Rolling = ground slope between 2 percent to 10 percent
- Hilly = ground slope greater than 10 percent

Oregon Department of Transportation Hydraulics Manual

https://www.oregon.gov/ODOT/GeoEnvironmental/Docs_Hydraulics_Manual/Hydraulics-07-F.pdf

Cover description

Cover type and hydrologic condition

Curve numbers for hydrologic soil group

A B C D

Open space (lawns, parks, golf courses, cemeteries, etc.)³:

Poor condition (grass cover < 50%)

68 79 86 89

Fair condition (grass cover 50% to 75%)

49 69 79 84

Good condition (grass cover > 75%)

39 61 74 80

Impervious areas:

Paved parking lots, roofs, driveways, etc. (excluding right-of-way)

98 98 98 98

Streets and roads:

Paved; curbs and storm sewers (excluding right-of-way)

98 98 98 98

Paved; open ditches (including right-of-way)

83 89 92 93

Gravel (including right-of-way)

76 85 89 91

Dirt (including right-of-way)

72 82 87 89

Urban districts:

Commercial and business

89 92 94 95

Industrial

81 88 91 93

Developing urban areas:

Newly graded areas (pervious areas only, no vegetation)⁵

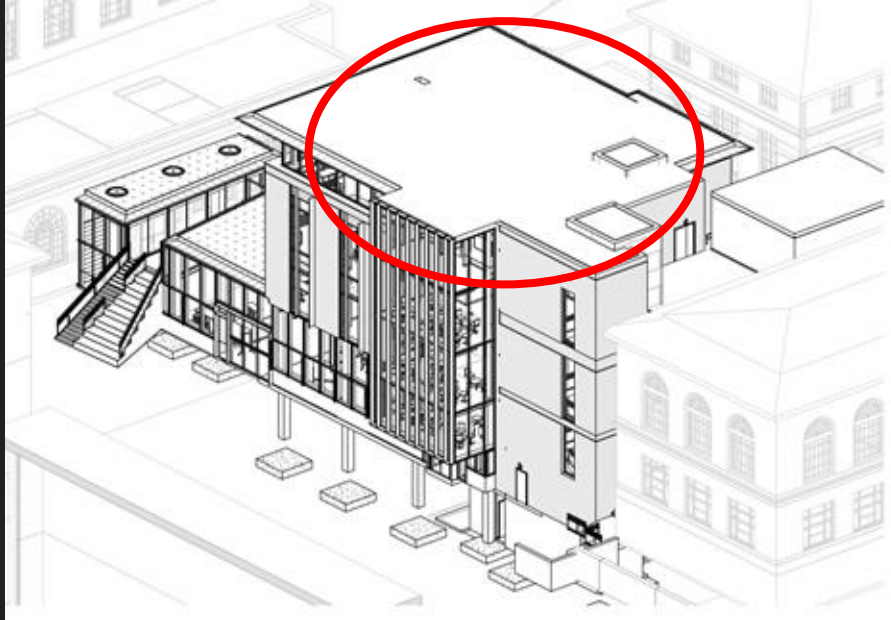
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Army Corps of Engineers Hydraulic Engineering Center

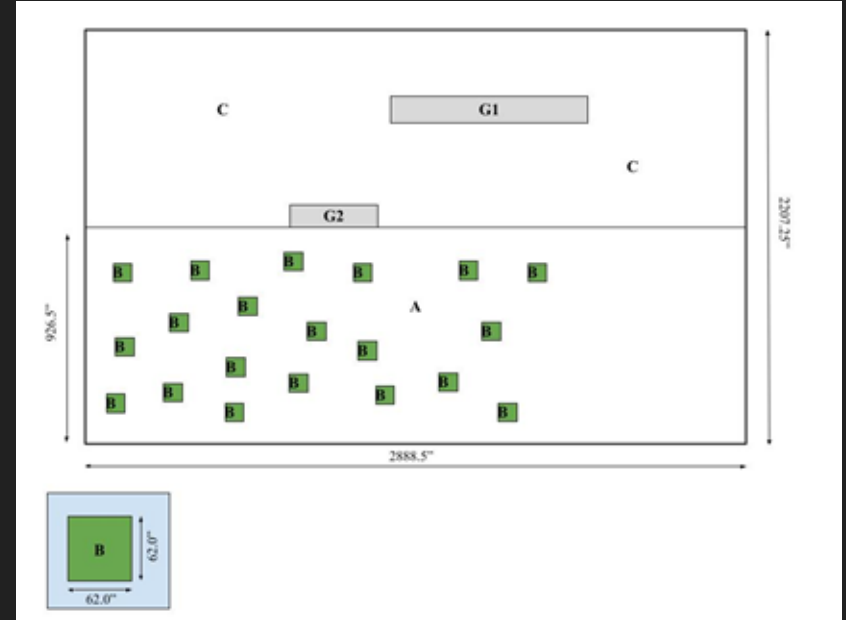
<https://www.hec.usace.army.mil/confluence/hmsdocs/hmstrm/cn-tables>

Design 1:

Implementing “Green Baskets” on the ANSYS Hall Roof



(Jackson, 2017)



Design 1:

continued

Ranking (Scale of 1 to 5)	Control Stormwater Runoff	Safety	Sustainability	Cost-Effectiveness	Building Integration
Design 1 (Andy)	3	5	5	5	5

Pros

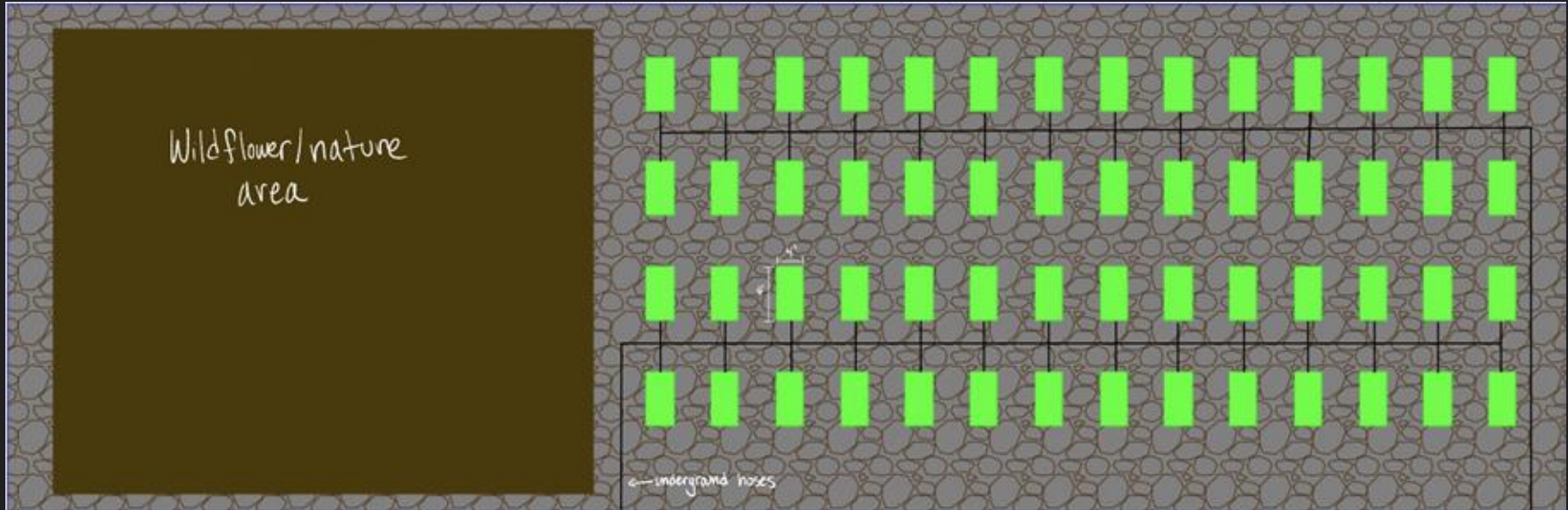
1. Reduce heat island effect
2. Low maintenance cost

Cons

1. Lack of performance
2. Difficulties in accessing the roof

Design 2:

Replacing concrete with garden areas and mulch/gravel ground



Design 2:

continued

Ranking (Scale of 1 to 5)	Control Stormwater Runoff	Safety	Sustainability	Cost-Effectiveness	Building Integration
Design 2 (Flannery)	4	5	5	3	3

Pros

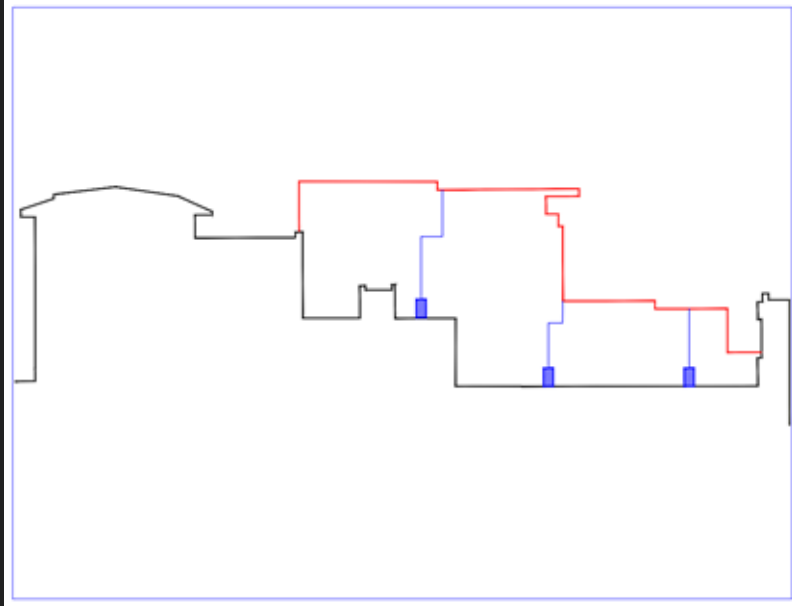
1. Decrease runoff rate by 50%
2. Sustainable
3. Safe

Cons

1. High cost
2. Assumptions may vary outcomes

Design 2:

Adding Rain Basins



Sketch of ANSYS Hall's east elevation



Four rain basins will be located on the ground level of ANSYS Hall

Design 3:

continued

Ranking (Scale of 1 to 5)	Control Stormwater Runoff	Safety	Sustainability	Cost-Effectiveness	Building Integration
Design 3 (Maddy)	3	4	5	4	1

Pros

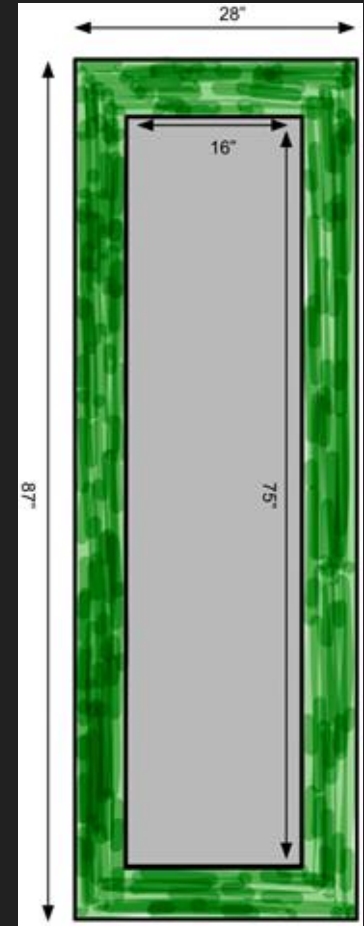
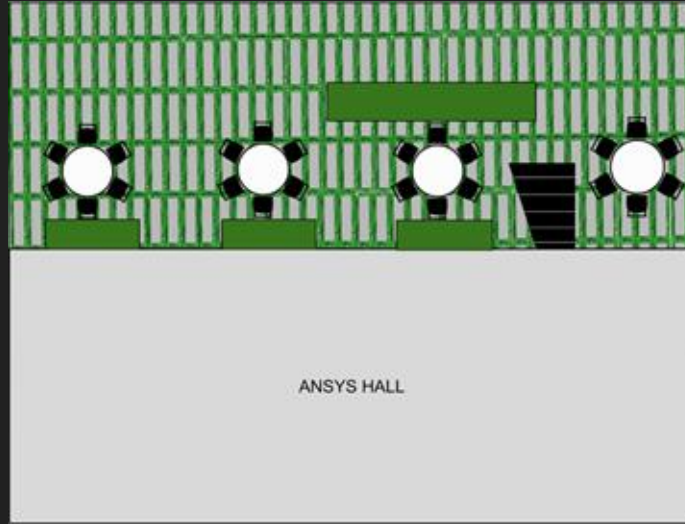
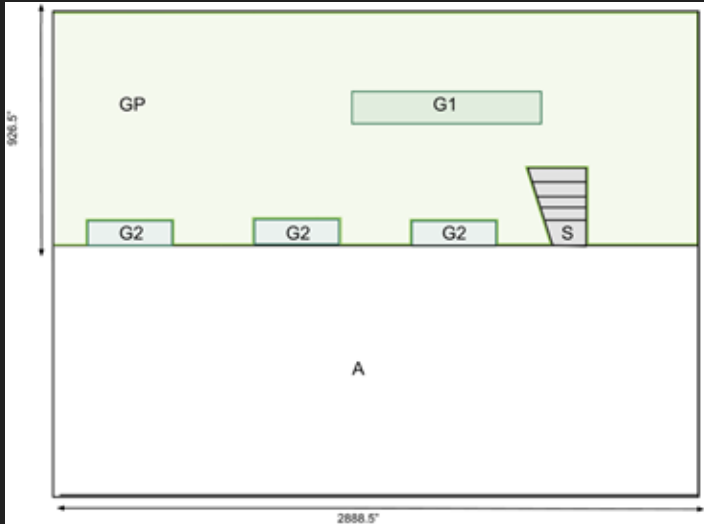
1. Direct storing of water
2. Low maintenance

Cons

1. Basins do not collect all stormwater
1. Poor building integration

Design 4:

Creating an interactive urban green space



Design 4:

continued

Ranking (Scale of 1 to 5)	Control Stormwater Runoff	Safety	Sustainability	Cost-Effectiveness	Building Integration
Design 3 (Krishnav)	4	5	5	4	4

Pros

1. Sustainable
2. Beautification of ANSYS Hall
3. Creating interactive space

Cons

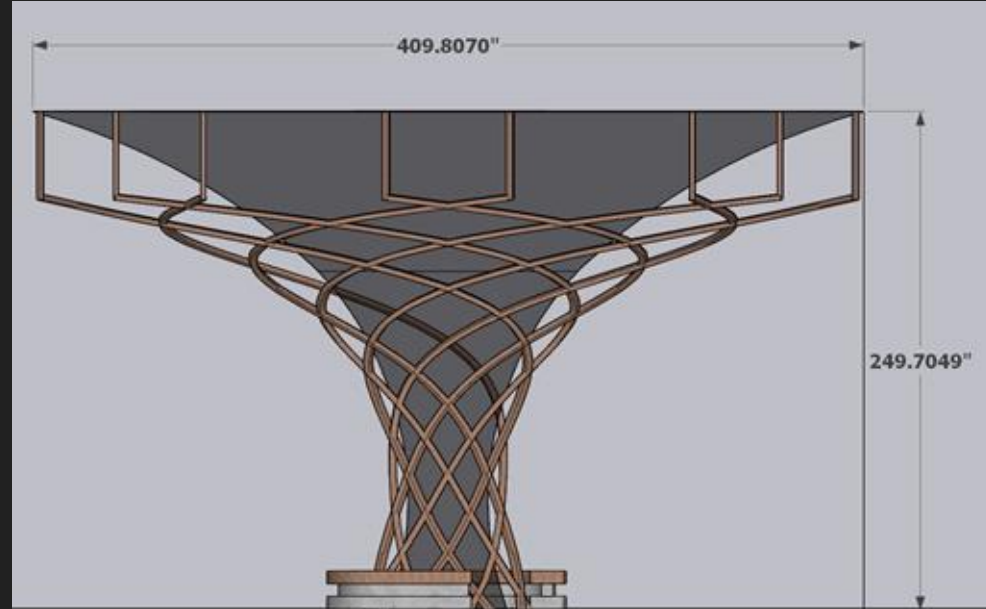
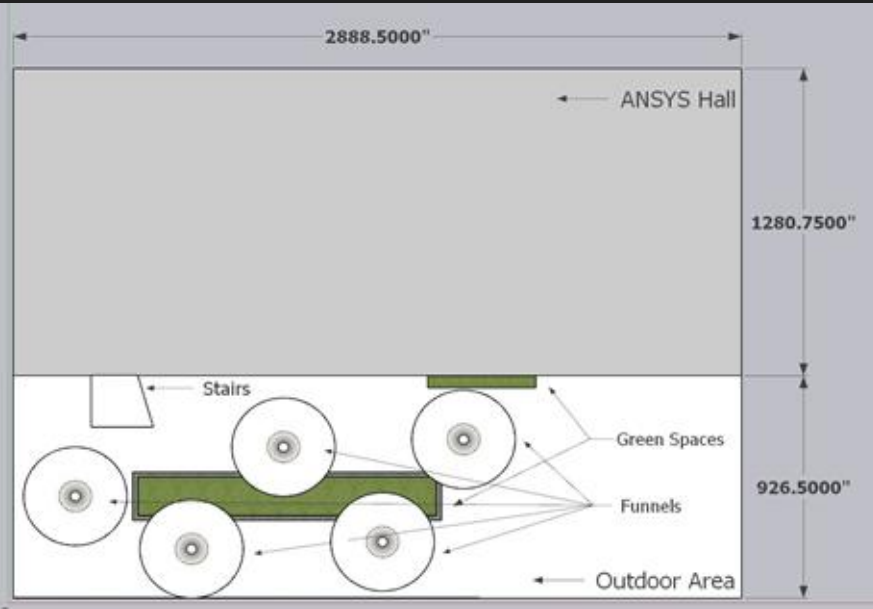
1. Costs may vary depending on the design
1. Required a lot of maintenance

Design 5:

Implementing “Rain Funnels” in front of ANSYS Hall



Design 5: *continued*



Design 5: *continued*

Ranking (Scale of 1 to 5)	Control Stormwater Runoff	Safety	Sustainability	Cost-Effectiveness	Building Integration
Design 5 (Emiliano)	2	4	3	3	4

Pros

1. Beautification the ANSYS Hall
2. Recreational functions

Cons

1. High cost
2. Risky operation
3. Lack of performance

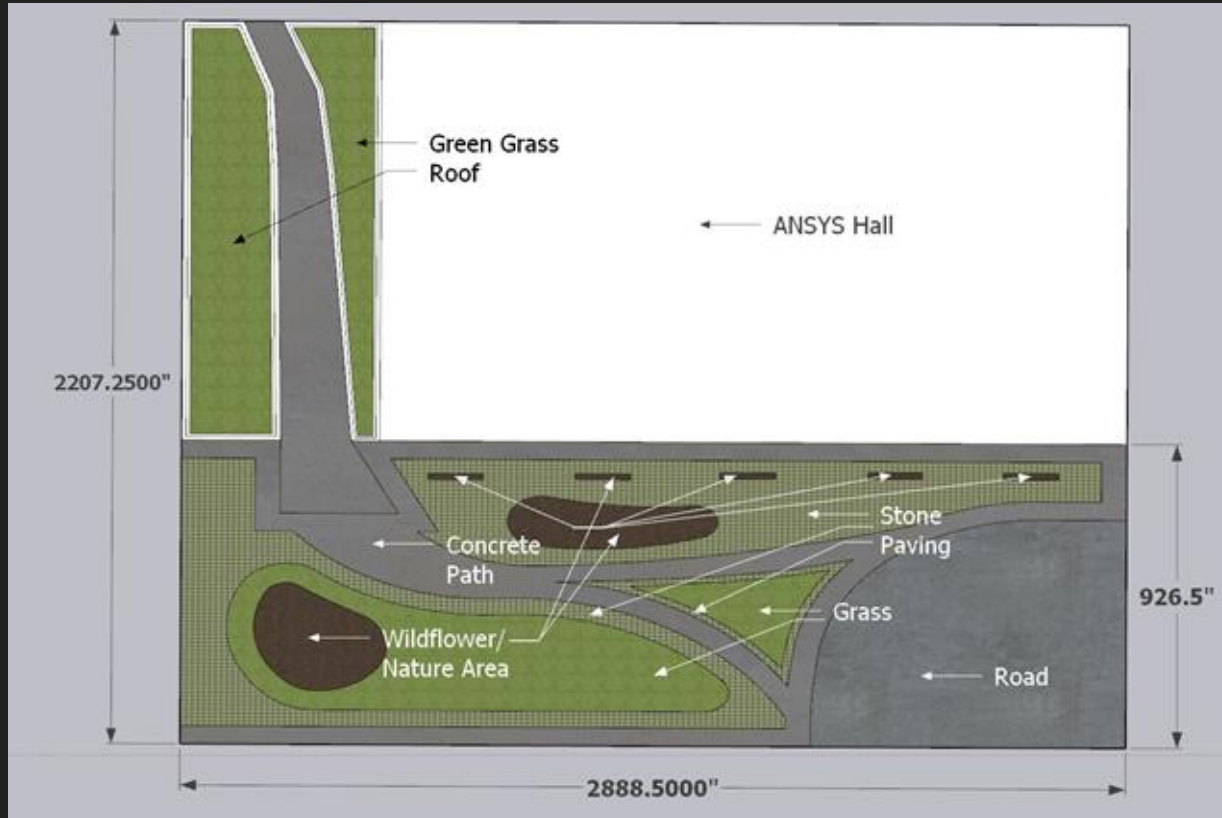
Comparing the Calculations among All Designs

	Runoff Rate (2 yr) (ft^3/sec)	Volume (2 yr) (in^3)	Runoff Rate (100 yr) (ft^3/sec)	Volume (100 yr) (in^3)
Design 1 (Andy)	0.0971	1.46×10^7	0.1852	2.90×10^7
Design 2 (Flannery)	0.0311	6.41×10^6	0.0621	1.49×10^7
Design 3 (Maddy)	0.0975	1.04×10^7	0.1884	2.54×10^7
Design 4 (Krishnav)	0.08323	1.28×10^7	0.1563	2.77×10^7
Design 5 (Emiliano)	0.08745	1.32×10^7	0.1652	2.65×10^7

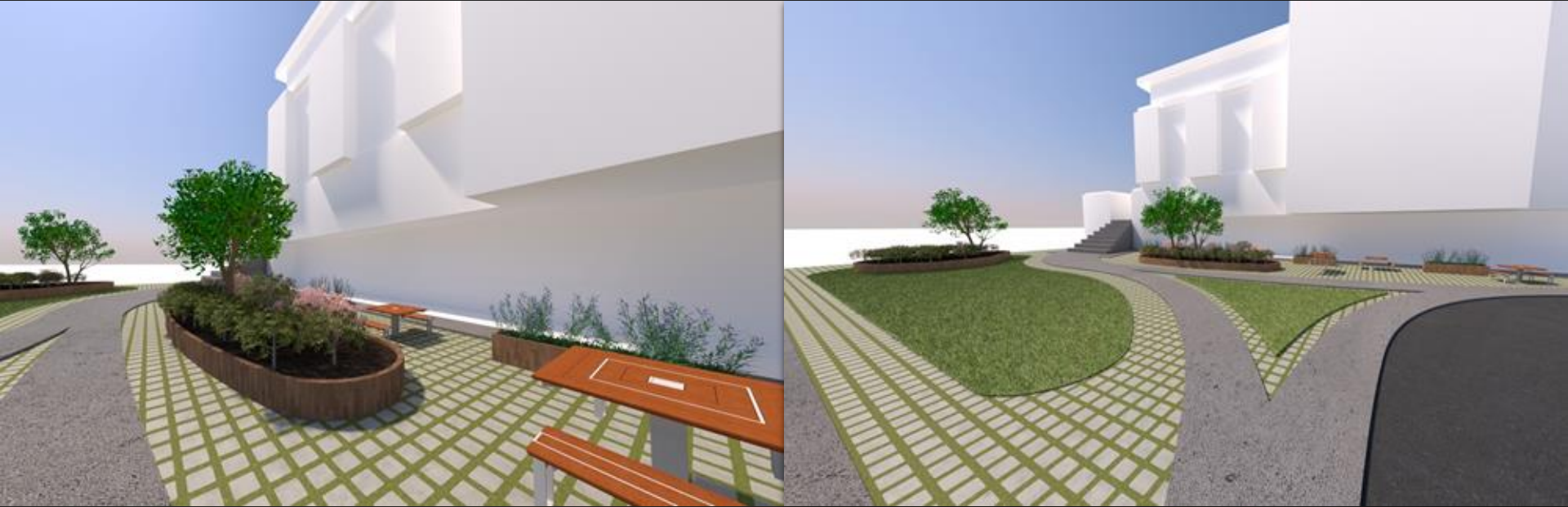
Comparing the Metrics among All Designs

Ranking (Scale of 1 to 5)	Control Stormwater Runoff	Safety	Sustainability	Cost-Effectiveness	Building Integration
Design 1 (Andy)	3	5	5	4	5
Design 2 (Flannery)	4	5	5	3	3
Design 3 (Maddy)	3	4	5	4	1
Design 4 (Krishnav)	4	5	5	4	4
Design 5 (Emiliano)	2	4	3	3	4

The Final Design Will Incorporate Both Green Roofs and Urban Green Space



The Final Design Will Provide More Space for People to Interact



The Final Design Has the Highest Metrics Score

Ranking (Scale of 1 to 5)	Control Stormwater Runoff	Safety	Sustainability	Cost-Effectiveness	Building Integration
Design 1 (Andy)	3	5	5	5	5
Design 2 (Flannery)	4	5	5	3	3
Design 3 (Maddy)	3	4	5	4	1
Design 4 (Krishnav)	4	5	5	4	4
Design 5 (Emiliano)	2	4	3	3	4
Final Design	5	5	5	4	5

The Final Design Is Effective in Reducing Stormwater

	Runoff Rate (2 yr) (ft^3/sec)	Volume (2 yr) (in^3)	Runoff Rate (100 yr) (ft^3/sec)	Volume (100 yr) (in^3)
Design 1 (Andy)	0.0971	1.46×10^7	0.1852	2.90×10^7
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Final Design	0.08237	1.27×10^7	0.1562	2.67×10^7

The Interactive Green Space Is the Most Functional and Cost Effective of Our Initial Designs

Construction Phase	Peak Rate (2-year storm) (ft^3/sec)	Peak Rate (100-year storm) (ft^3/sec)	Volume (2-year storm) (in^3)	Volume (100-year storm) (in^3)
Pre-Construction	0.085	0.16	1.3×10^7	2.72×10^7
Post-Construction	0.098	0.187	1.52×10^7	3.00×10^7
Final Design	0.082	0.156	1.27×10^7	2.67×10^7
Difference Between Post-Construction and Final Design	0.016	0.031	2.5×10^6	0.33×10^7
Difference Between Pre-Construction and Final Design	0.003	0.004	3×10^5	5×10^5

Q&A



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